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P1. $R_S(F) = E_\varepsilon[\sup_{f\in F} \frac{1}{n}\sum_{i=1}^n \varepsilon_i f(x_i)]$

$= E_\varepsilon[\sup_{f\in F} \frac{1}{n}\sum_{i=1}^n \varepsilon_i w^T x_i]$

$= E_\varepsilon[\sup_{f\in F} \frac{1}{n} w^T \sum_{i=1}^n \varepsilon_i x_i]$ (Let $\sum_{i=1}^n \varepsilon_i x_i = v$)

$= E_\varepsilon[\sup_{f\in F} \frac{1}{n} w^T v]$

For $\sup_{f\in F} \frac{1}{n} w^T v$. we know $\frac{1}{n} w^T v \le \frac{1}{n}|w^T v| \le \frac{1}{n}\|w^T\|_1 \|v\|_\infty \le \frac{1}{n} a \|v\|_\infty$

so $\sup_{\|w\|_1 \le a} \frac{1}{n} w^T v = \frac{a}{n}\|v\|_\infty$

$\therefore R_S(F) = \frac{a}{n} E_\varepsilon[\max_{i\in\{1,\dots,d\}} |\sum_{i=1}^n \sigma_i x_i^{(i)}|]$

$= \frac{a}{n} E_\varepsilon[\max_{i\in\{1,\dots,d\}} s\cdot \sum_{i=1}^n \sigma_i x_i^{(i)}]$

$= \frac{a}{n} E_\varepsilon[\max_{i\in\{1,\dots,d\}} \sum_{i=1}^n \sigma_i s\cdot x_i^{(i)}]$

Let $u_j = (x_1^{(j)}, x_2^{(j)}, \dots, x_n^{(j)})$

$V = (u_1, -u_1, u_2, -u_2, \dots, u_n, -u_n)$

$|V| = 2d$ $\|V\|_2 = \sqrt{\sum_{i=1}^n (x_i^{(i)})^2} \le \sqrt{\sum_{i=1}^n b^2} = b\sqrt{n}$

$\therefore R_S(F) = \frac{a}{n} E_\varepsilon[\max_{i\in(1,\dots,d)} \sum_{i=1}^n \sigma_i s x_i] \le \frac{a}{n} b\sqrt{n}\sqrt{2\log(2d)} = ab\sqrt{\frac{2\log(2d)}{n}}$

P2.

1. $S = \{x_1, \dots, x_k, \dots, x_n\}$

$S' = \{x_1, \dots, x_k', \dots, x_n\}$

The U-statistic is defined as $U_n(S) = \frac{1}{n(n-1)} \sum_{i=1}^n \sum_{j\ne i} g(x_i, x_j)$

so $U_n(S) - U_n(S') = \frac{1}{n(n-1)}[\sum_{j\ne k}(g(x_k,x_j) - g(x_k',x_j)) + \sum_{i\ne k}(g(x_i,x_k) - g(x_i,x_k'))]$

$\because |g(x,x')| \le b$ $\therefore |g(x_k,x_j) - g(x_k',x_j)| \le |g(x_k,x_j)| + |g(x_k',x_j)| \le 2b$

$\therefore |U_n(S) - U_n(S')| \le \frac{1}{n(n-1)}[(n-1)\cdot 2b + (n-1)\cdot 2b] = \frac{4b}{n}$

$\therefore c_i = \frac{4b}{n}$ for all $i = 1, \dots, n$

2. From McDiarmid's Inequality

$U_n \le E[U_n] + \sqrt{\frac{\log(1/\delta)}{2}\sum_{i=1}^n c_i^2} \le E[U_n] + \sqrt{\frac{\log(1/\delta)}{2} n(\frac{4b}{n})^2}$

$= E[U_n] + 4b\sqrt{\frac{\log(1/\delta)}{2n}}$

$\therefore U_n - E[U_n] \le 4b\sqrt{\frac{\log(1/\delta)}{2n}}$

P3. $f(z) = \frac{1}{2}(f_1(z) + f_2(z) + |f_1(z) - f_2(z)|)$

$R_S(F) = E_\varepsilon[\sup_{f\in F} \frac{1}{n}\sum_{i=1}^n \varepsilon_i f(z_i)]$

$= E_\varepsilon[\sup_{f_1\in F_1, f_2\in F_2} \frac{1}{n}\sum_{i=1}^n \varepsilon_i \frac{1}{2}(f_1(z_i) + f_2(z_i) + |f_1(z_i) - f_2(z_i)|)]$

$\le E_\varepsilon[\sup_{f_1\in F_1} \frac{1}{n}\sum_{i=1}^n \varepsilon_i \frac{1}{2} f_1(z_i) + \sup_{f_2\in F_2} \frac{1}{n}\sum_{i=1}^n \varepsilon_i \frac{1}{2} f_2(z_i) + \sup_{f_1\in F_1, f_2\in F_2} \frac{1}{n}\sum_{i=1}^n \varepsilon_i \frac{1}{2}|f_1(z_i) - f_2(z_i)|]$

$\le \frac{1}{2}R_S(F_1) + \frac{1}{2}R_S(F_2) + \frac{1}{2}E_\varepsilon[\sup_{f_1\in F_1, f_2\in F_2} \frac{1}{n}\sum_{i=1}^n \varepsilon_i |f_1(z_i) - f_2(z_i)|]$

Let $\phi(x) = |x|$ $\because |x| = |x - y + y| \le |x-y| + |y|$ $\Rightarrow |x| - |y| \le |x-y|$

$|y| = |y - x + x| \Rightarrow |y| - |x| \le |y-x| = |x-y|$

$\therefore ||x| - |y|| \le |x-y|$ $\phi(x)$ is a 1-Lipschitz

$\therefore E_\varepsilon[\sup_{f_1,f_2} \frac{1}{n}\sum_{i=1}^n \varepsilon_i |f_1(z_i) - f_2(z_i)|] \le E_\varepsilon[\sup_{f_1,f_2} \frac{1}{n}\sum_{i=1}^n \varepsilon_i (f_1(z_i) - f_2(z_i))]$

$\le E_\varepsilon[\sup_{f_1} \frac{1}{n}\sum_{i=1}^n \varepsilon_i f_1(z_i) + \sup_{f_2} \frac{1}{n}\sum_{i=1}^n \varepsilon_i f_2(z_i)]$

$\le R_S(F_1) + R_S(F_2)$

$\therefore R_S(F) \le \frac{1}{2}R_S(F_1) + \frac{1}{2}R_S(F_2) + \frac{1}{2}R_S(F_1) + \frac{1}{2}R_S(F_2)$

$\le R_S(F_1) + R_S(F_2)$

P4. $R_S(H) = E_\sigma[\sup_{\|w\|_2\le 1} \frac{1}{n}\sum_{i=1}^n \sigma_i \max\{0, w^T x_i\}]$

$\phi(x) = \max\{0, x\}$ is 1-Lipschitz function

so $R_S(H) \le R_S(G) = E_\sigma[\sup_{\|w\|_2\le 1} \frac{1}{n}\sum_{i=1}^n \sigma_i w^T x_i]$

$= \frac{1}{n} E_\sigma[\sup_{\|w\|_2\le 1} w^T \sum_{i=1}^n \sigma_i x_i]$

$\le \frac{1}{n} E_\sigma[\sup_{\|w\|_2\le 1} \|w^T\|_2 \cdot \|\sum_{i=1}^n \sigma_i x_i\|_2]$

$\le \frac{1}{n} E_\sigma[\|\sum_{i=1}^n \sigma_i x_i\|_2]$

$\because E[\sqrt{x}] \le \sqrt{E[x]}$

$\therefore \frac{1}{n} E_\sigma[\|\sum_{i=1}^n \sigma_i x_i\|_2] \le \frac{1}{n}(E_\sigma[\|\sum_{i=1}^n \sigma_i x_i\|_2^2])^{\frac{1}{2}}$

$= \frac{1}{n}[E_\sigma[\sum_{i=1}^n \sum_{j=1}^n \sigma_i\sigma_j (x_i^T x_j)]]^{\frac{1}{2}}$

$= \frac{1}{n}(E_\sigma[\sum_{i=1}^n \sigma_i^2 x_i^2])^{\frac{1}{2}}$ ($E_\sigma(\sigma_i\sigma_j) = 0$)

$= \frac{1}{n}(\sum_{i=1}^n \|x_i\|_2^2 E_\sigma[\sigma_i^2])^{\frac{1}{2}}$

$= \frac{1}{n}(\sum_{i=1}^n \|x_i\|_2^2)^{\frac{1}{2}}$ ($E_\sigma(\sigma_i^2) = 1$)

$\therefore R_S(H) \le \frac{1}{n}(\sum_{i=1}^n \|x_i\|_2^2)^{\frac{1}{2}}$

P5

(1). $g(s) = \|\frac{1}{n}\sum_{i=1}^n \phi(z_i) - E[\phi(z)]\|_2$

$= \|\frac{1}{n}\sum_{i=1}^n \phi(z_i) - E_{s'}[\frac{1}{n}\sum_{i=1}^n \phi(z_i')]\|_2$

$= \|E_{s'}[\frac{1}{n}\sum_{i=1}^n (\phi(z_i) - \phi(z_i'))]\|_2$

$E[g(s)] = E_s\|E_{s'}[\frac{1}{n}\sum_{i=1}^n (\phi(z_i) - \phi(z_i'))]\|_2$

$\le E_{s,s'}\|\frac{1}{n}\sum_{i=1}^n (\phi(z_i) - \phi(z_i'))\|_2$ (Apply $\|E(x)\| \le E(\|x\|)$)

$= E_{s,s',\varepsilon}\|\frac{1}{n}\sum_{i=1}^n \varepsilon_i(\phi(z_i) - \phi(z_i'))\|_2$ (Introduce Rademacher random variables)

$\le E_{s,\varepsilon}\|\frac{1}{n}\sum_{i=1}^n \varepsilon_i \phi(z_i)\|_2 + E_{s',\varepsilon}\|\frac{1}{n}\sum_{i=1}^n \varepsilon_i \phi(z_i')\|_2$ (Triangle Inequality)

$= \frac{2}{n} E_{s,\varepsilon}\|\sum_{i=1}^n \varepsilon_i \phi(z_i)\|_2$

$\le \frac{2}{n}\sqrt{E_{s,\varepsilon}\|\sum_{i=1}^n \varepsilon_i \phi(z_i)\|_2^2}$ ($E(X^2) \le \sqrt{E(X)}$)

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$= \frac{2}{n}\sqrt{E_{S,\varepsilon}\left(\sum_{i=1}^{n}\sum_{j=1}^{n}\varepsilon_i\varepsilon_j\phi(z_i)\phi(z_j)\right)}$

$= \frac{2}{n}\sqrt{E_{S,\varepsilon}\left[\sum_{i=1}^{n}\phi(z_i)^2\right]}$ $\quad (E(\varepsilon_i\varepsilon_j)=0,\ E(\varepsilon_i^2)=1)$

$\le \frac{2}{n}\sqrt{n\,B^2}$ $\quad (\|\phi(z)\| \le B$

$= \frac{2B}{\sqrt{n}}$

2. $S = \{z_1, z_2, \dots, z_k, \dots, z_n\}$

$S' = \{z_1, z_2, \dots z_k', \dots, z_n\}$

$|g(S) - g(S')|$

$= \left| \|\frac{1}{n}\sum_{i=1}^{n}\phi(z_i) - \mu\|_2 - \|\frac{1}{n}\sum_{i=1}^{n}\phi(z_i') - \mu\|_2 \right|$

$\le \left| \|\frac{1}{n}\sum_{i=1}^{n}\phi(z_i) - \mu - \frac{1}{n}\sum_{i=1}^{n}\phi(z_i') + \mu\|_2 \right|$ $\quad (|\|x\| - \|y\|| \le |x-y|)$

$= \|\frac{1}{n}\sum_{i=1}^{n}(\phi(z_i) - \phi(z_i'))\|_2$

$= \|\frac{1}{n}(\phi(z_k) - \phi(z_k'))\|_2$

$\le \|\frac{1}{n}\phi(z_k)\| + \|\frac{1}{n}\phi(z_k')\|$

$= \frac{1}{n}\|\phi(z_k)\| + \frac{1}{n}\|\phi(z_k')\|$

$\le \frac{1}{n}(B + B) = \frac{2B}{n}$ $\quad \therefore c_i = \frac{2B}{n}$ For all $i = 1, 2, \dots, n$

3. From McDiarmid Inequality

We know $P(g(S) - E(g(S)) \ge \left(\frac{\log(1/\delta)}{2}\sum_{i=1}^{n}c_i^2\right)^{\frac{1}{2}}) \le \delta$

$\because \sum_{i=1}^{n}c_i^2 = \frac{4B^2}{n^2}\cdot n = \frac{4B^2}{n}$

$\therefore \left(\frac{\log(1/\delta)}{2}\frac{4B^2}{n}\right)^{\frac{1}{2}} = \frac{B}{\sqrt{n}}\cdot\sqrt{2\log(1/\delta)}$

$\because E(g(S)) \le \frac{2B}{\sqrt{n}}$

$\therefore E(g(S)) + \frac{B}{\sqrt{n}}\cdot\sqrt{2\log(1/\delta)} \le \frac{B}{\sqrt{n}}(2 + \sqrt{2\log(1/\delta)})$

$\therefore P\left(g(S) > \frac{B}{\sqrt{n}}(2 + \sqrt{\log(1/\delta)})\right) \le \delta$

P6.

1. From Massart Lemma

$E\left[\sup_{a\in A}\frac{1}{n}\sum_{i=1}^{n}\varepsilon_i a_i\right] \le \frac{1}{n}\max_{a\in A}\|a\|_2\sqrt{2\log|A|}$

$a_w = (\langle w, x_1\rangle, \langle w, x_2\rangle, \dots \langle w, x_n\rangle)$

$|A| = 2^d$ $\quad (w \in \{+1, -1\}$

$\langle w, x_i\rangle \le \|w\|_\infty \|x_i\|_1 \le r$

$\therefore \|a_w\|_2 = \sqrt{\sum_{i=1}^{n}\langle w, x_i\rangle^2} \le \sqrt{nr^2} = r\sqrt{n}$

$\therefore E\left[\sup_{a\in A}\frac{1}{n}\sum_{i=1}^{n}\varepsilon_i a_i\right] \le \frac{1}{n}\max_{a\in A}\|a\|_2\sqrt{2\log|A|}$

$= \frac{1}{n}\, r\sqrt{n}\sqrt{2d\log 2}$

$E\left[\sup_{a\in A}\sum_{i=1}^{n}\varepsilon_i a_i\right] \le r\sqrt{n}\sqrt{2d\log 2}$

2. $R(S) = E_\sigma\left[\sup_{\|w\|_1\le 1}\frac{1}{n}\sum_{i=1}^{n}\sigma_i w^T x\right]$

$= E_\sigma\left[\sup_{\|w\|_1\le 1} w^T\frac{1}{n}\sum_{i=1}^{n}\sigma_i x\right]$

$\le E_\sigma\left[\sup_{\|w\|_1\le 1}\|w\|_\infty\left\|\frac{1}{n}\sum_{i=1}^{n}\sigma_i x\right\|_1\right]$

$= \frac{1}{n}E_\sigma\left[\left\|\sum_{i=1}^{n}\sigma_i x\right\|_1\right]$

$\le \frac{r\sqrt{n}}{n}(2d\log 2)^{\frac{1}{2}}$ $\quad$ (From problem 1.)

$= \frac{r(2d\log 2)^{\frac{1}{2}}}{\sqrt{n}}$

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